

# llmXive Automated Review of Search Beyond What Can Be Taught: Evolving the Knowledge Boundary in Agentic Visual Generation

*llmXive automated review panel*

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**Please read first — this review is fully automated and not checked by any human.** It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

## How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

**This paper's panel (9 lenses):** Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

## Claim Accuracy

*Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a compelling framework for agentic visual generation, but there are specific instances where the cited evidence or model names do not align perfectly with the claims, requiring clarification to ensure the reader can trust the reported results.

First, the evaluation protocol relies heavily on a “Gemini-3-Flash” judge (Section 3.1, Appendix). The bibliography entry `nanobanana2025gemini` lists “Gemini 2.5 Flash / Gemini 3 Pro” but the text consistently refers to “Gemini-3-Flash” as a specific model. Given that “Gemini 3” is a future-dated or non-standard name in the public record (as of the paper’s 2026 context, models are typically 1.5 or 2.0), this raises a concern about whether the model name is a hallucination or a typo for “Gemini 2.5 Flash”. Since the entire evaluation metric (Spearman  $\rho = 0.87$ ) and the baseline comparisons depend on this specific judge, the exact model identity must be verified and corrected if it is a hallucinated name.

Second, there is a minor inconsistency in model naming between the text and tables. Section 3.1 refers to “Qwen-Image-1” dropping nearly 40 points, while Table 1 lists “Qwen-Image” (NoSearch 67.4, Search 27.9, drop 39.5). The bibliography has a single entry `qwenimage` (2025). The text should clarify if “Qwen-Image” in the table is indeed “Qwen-Image-1” to ensure the 40-point claim is attributed to the correct model version, especially since “Qwen-Image-2” is also mentioned as a separate commercial baseline.

Finally, while the numerical claims in Section 4.1 (Phase 2 exceeding Oracle by 0.6 points) are mathematically consistent with Table 2, the phrasing “slightly exceeding” for a 0.6 point difference on a 100-point scale without reported statistical significance (e.g., standard deviation or p-value) is a minor overstatement of the result’s robustness. However, the primary issue remains the potential hallucination of the “Gemini-3-Flash” model name, which undermines the reproducibility of the evaluation.

### Action items.

- **[writing]** Section 3.1 cites ‘Gemini-3-Flash’ as the judge. The bib entry ‘nanobanana2025gemini’ lists ‘Gemini 2.5 Flash / Gemini 3 Pro’. Verify if ‘Gemini-3-Flash’ is a hallucinated name or a typo for ‘Gemini 2.5 Flash’, as the specific model identity is critical for the reported 0.87 correlation.
- **[writing]** Section 3.1 claims ‘Qwen-Image-1 drops nearly 40 points’, but Table 1 lists ‘Qwen-Image’ (NoSearch 67.4, Search 27.9). Clarify if ‘Qwen-Image’ in the table corresponds to ‘Qwen-Image-1’ in the text to ensure the 40-point claim is supported by the correct model identifier.
- **[writing]** Section 4.1 states Phase 2 ‘slightly exceeding’ the Oracle (31.8 vs 31.2). While mathematically correct, claiming a 0.6 point gain as ‘exceeding’ without statistical significance (e.g., p-value or std dev) is an overstatement of the result’s robustness. Qualify the claim or

add statistical context.

## Figure Critic

*Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_figure\_critic.md

### Figure 1

The figure presents a heatmap of percentages but fails to define the metric, rendering the caption’s claim of ‘degradation’ unsupported. Additionally, the colorbar scale does not accommodate the maximum data value (65%), and the caption contains a missing reference.

### Figure 2

The figure is a black-and-white photograph of women carrying baskets that bears no relation to the caption’s description of an AI visual generation workflow, rendering the figure completely irrelevant to the paper’s claims.

### Figure 3

The rendered image is a generic collage of AI art that fails to illustrate the specific ‘search-augmented generations’ or ‘failure categories’ described in the caption. The caption itself is grammatically broken and contains missing references, making the figure and its description unintelligible.

### Figure 4

Figure 4 effectively illustrates the two paradigms for visual generation described in the caption. The diagram is clear, with distinct sections for ‘Prompt Rewrite’ and ‘SearchGen (Ours)’, and the flow of information is easy to follow. All components, including the LLM, Agent, Search Tool, Web Corpus, and Generator, are appropriately labeled and connected.

### Figure 5

The figure effectively visualizes the language distribution and prompt length differences, but the caption contains a critical typo omitting the dataset name, and the chart lacks a source label for the data.

## Figure 6

The treemap visualizes domain distribution but suffers from a confusing hierarchical structure where the largest blocks are undefined, and the caption contains a broken cross-reference to an appendix figure.

## Figure 7

Figure 7 presents entity frequency and knowledge gap distributions, but the histogram in panel (b) visually contradicts the caption’s claim of a mean of 5.2 gaps, and the cumulative percentage line lacks an in-figure legend.

## Figure 8

The figure effectively visualizes the performance collapse across generators, but the axis labels are cluttered and the caption’s specific numerical claims (-0.1, -39.1) do not match the rounded values shown on the chart (-0, -39).

## Figure 9

The figure provides a clear visual overview of the co-training framework, but the caption is truncated mid-sentence, and the feedback loop for rejection finetuning lacks clear directional flow.

## Figure 10

Figure 10 contains a significant contradiction where the caption claims monotonic improvement across tiers, while the data shows performance drops on harder tiers (Set III). Additionally, the caption fails to mention the Bagel-7B results explicitly shown in panel (a).

**Action items.**

- **[science]** Figure 1: The caption claims the heatmap shows ‘knowledge-driven degradation’ and ‘failure modes,’ but the cells contain positive percentages (e.g., 65%, 42%) without a defined baseline or metric. It is unclear if these values represent failure rates, entity frequencies, or another metric, making the claim of ‘degradation’ unsupported by the visual data.
- **[writing]** Figure 1: The caption contains a dangling reference ‘in .’ where a dataset or section name is missing, leaving the context of the data undefined.
- **[science]** Figure 1: The colorbar scale (0–60+) contradicts the data labels (percentages). The cell labeled ‘65%’ is colored dark red, but the colorbar’s maximum tick is 60, and the color mapping for values >60 is undefined.
- **[fatal]** Figure 2: The rendered image is a black-and-white photograph of women carrying baskets in a field, which completely contradicts the caption’s description of ‘Yang Chaoyue request’ references and ‘end-to-end example’s visual output’ for an AI generation system.
- **[science]** Figure 2: The image fails to support the paper’s claims about agentic visual

generation, as it shows a historical photograph rather than the described ‘retrieved references’ (scene, costume, likeness) or the ‘generated image’.

- **[fatal]** Figure 3: The rendered image is a collage of unrelated AI-generated images (portraits, anime, diagrams) that does not match the caption’s claim of showing ‘representative search-augmented generations’ or ‘failure categories’.
- **[science]** Figure 3: The image contains no labels, annotations, or visual indicators to identify the ‘twelve failure categories’ mentioned in the caption, making the figure impossible to interpret.
- **[writing]** Figure 3: The caption contains grammatical errors and missing references (e.g., ‘from ,’, ‘captures the production-scale’) that render the text incoherent.
- **[fatal]** Figure 5: The caption contains a missing dataset name (‘distribution of .’) and the chart labels use ‘Chinese’ and ‘English’ without specifying the dataset source, making the figure contextually incomplete.
- **[science]** Figure 6: The caption states ‘area reflects relative prompt counts,’ but the treemap is dominated by two massive blocks (‘People & Professions’ and ‘Screen & Performance Media’) that are not defined as categories in the smaller blocks, creating a confusing hierarchy where the visual ‘mass’ does not clearly map to the ‘domain categories’ mentioned.
- **[writing]** Figure 6: The caption contains a broken cross-reference (‘deferred to Appendix (Figure )’) with a missing figure number, making it impossible to verify the claim about ‘uniform severity’ mentioned in the text.
- **[science]** Figure 7(b): The caption states the mean number of knowledge gaps is 5.2, but the histogram bars are centered on bins (0-2, 3-5, 6-8, 9-12, 13+) where the highest frequency is in the 3-5 bin. A mean of 5.2 would imply a distribution skewed heavily towards the higher bins (6-8, 9-12, 13+), yet the bar for 6-8 is significantly lower than 3-5, and the bars for 9-12 and 13+ are very small. The visual distribution contradicts the stated mean of 5.2.
- **[writing]** Figure 7(b): The right y-axis label ‘Cumulative %’ is present, but the red line plot representing the cumulative percentage lacks a clear legend entry within the figure itself to distinguish it from the blue bars, relying solely on the caption for interpretation.
- **[science]** Figure 8: The caption claims drop magnitudes range from -0.1 to -39.1, but the chart labels show -0 (GPT-Image-2) and -39 (Qwen-2), creating a discrepancy between the text and the visual data.
- **[science]** Figure 8: The x-axis labels are slanted and overlap significantly (e.g., ‘Bagel’, ‘Klein-4B’, ‘Klein-9B’), making them difficult to read and potentially illegible in lower resolutions.
- **[writing]** Figure 8: The y-axis label ‘Score (0-100)’ is present, but the specific metric being averaged (e.g., ‘93.1% of unique visual entities’ or a specific quality score) is not defined in the caption or axis.

- **[writing]** Figure 9: The caption text is truncated at the end (‘...reasoned sear’), cutting off the description of Phase 2.
- **[science]** Figure 9: The diagram shows a ‘Rejection Finetuning’ loop on the bottom right, but the arrow direction and connection to the ‘VLM Reasoner’ are ambiguous, making the feedback mechanism unclear.
- **[science]** Figure 10(a): The caption claims ‘All three tiers show monotonic improvement,’ but the Klein-4B chart shows a score drop from Set I (34.1) to Set III (27.4), and the Bagel-7B chart shows a drop from Set II (29.3) to Set III (22.6). The bars represent difficulty tiers, not a temporal progression of improvement.
- **[writing]** Figure 10(a): The caption states the bars show stages for ‘Klein-4B’, but the rendered figure explicitly contains a second panel for ‘Bagel-7B’ which is not mentioned in the caption text.
- **[writing]** Figure 10(b): The caption describes the shaded region as ‘newly internalized knowledge’ but does not explicitly define the shaded area in the plot (which visually represents the area between the base and DPO curves).

## Jargon Police

*Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_jargon\_police.md

The paper is generally well-written for an adjacent-field reader, with most acronyms (e.g., DPO, RFT, SFT) defined at first use or being standard in the field. However, there are several instances of undefined mathematical notation and specific model names that could stall a reader from a neighboring subfield (e.g., a computer vision researcher not deeply familiar with the specific flow-matching literature or the exact product codenames).

Specifically:

1. **Undefined Symbols in Equations:** In Section 3.2, the symbols  $\epsilon$  (tolerance),  $Q$  (quality function), and  $\mathbf{v}_i$  (velocity field) appear in equations or algorithm descriptions without explicit definitions in the surrounding prose. While the context implies their meaning, a rigorous definition (e.g., “where  $\epsilon$  is the tolerance threshold”) is missing at the point of first use.
2. **Product Codenames:** Terms like “Nano Banana Pro” and “Nano Banana” are used as model identifiers. While they are cited, a brief parenthetical clarification (e.g., “Gemini 3 Pro Image (referred to as Nano Banana Pro)”) would help a reader unfamiliar with the specific product naming conventions of the cited companies.
3. **Protocol Naming:** The “gate-filter-integrate” protocol is introduced as a solution to noise,

but the explicit naming of the three stages as a formal “protocol” could be slightly more prominent in the introductory sentence of Section 3.1 to ensure the reader immediately grasps the structure.

These are minor issues that can be resolved with simple parenthetical definitions or clarifying clauses, ensuring the paper is fully self-contained for a competent PhD from an adjacent field.

#### Action items.

- **[writing]** Undefined Symbols in Equations: In Section 3.2, the symbols  $\epsilon$  (tolerance),  $Q$  (quality function), and  $v_{\perp}$  (velocity field) appear in equations or algorithm descriptions without explicit definitions in the surrounding prose. While the context implies their meaning, a rigorous definition (e.g., “where  $\epsilon$  is the tolerance threshold”) is missing at the point of first use.

## Logical Consistency

*Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a logically coherent argument: it identifies a “world-knowledge bottleneck,” demonstrates that naive search exacerbates the problem, and proposes a co-training framework to discover a generator-specific “knowledge boundary.” The definitions of internalizable versus contextual knowledge are well-motivated by the empirical findings in Section 2, and the proposed solution follows naturally from the identified failure modes.

However, there are specific numerical inconsistencies regarding dataset sizes that break the internal consistency of the reported statistics.

First, the Abstract and Section 1 state the dataset contains **20,839** prompts. However, Section 2.2 states the training dataset comprises **20,188** rows and the test set contains **751** prompts. The sum ( $20,188 + 751$ ) is **20,939**, which contradicts the total of 20,839 cited in the Abstract. This 100-prompt discrepancy is unexplained and creates confusion about the exact scale of the benchmark.

Second, while the partitioning of the test set into “NoSearch” (100), “VisualSearch” (387), and “TextualSearch” (264) sums correctly to 751, the text in Section 2.3 refers to a “Search-Intensive” set of 651 prompts. The logic holds that  $387 + 264 = 651$ , but the paper should explicitly state that “Search-Intensive” is defined as the union of VisualSearch and TextualSearch to avoid any ambiguity about whether “Search-Intensive” is a separate category or a superset.

These are primarily numerical consistency issues that require a quick audit of the data split counts to ensure the numbers in the Abstract, Section 2.2, and Section 2.3 align perfectly. The logical flow of the argument remains intact, but the precision of the reported numbers needs correction.

#### Action items.



- **[writing]** Section 2.2 states the training set has 20,188 rows, but the Abstract claims 20,839 total prompts. With a 751-prompt test set,  $20,188 + 751 = 20,939$ , contradicting the 20,839 total. Clarify if the training count is 20,088 or if the total is 20,939 to ensure numerical consistency.
- **[writing]** Section 2.3 defines ‘Search-Intensive’ as 651 prompts, while Table 2 lists ‘VisualSearch’ (387) and ‘TextualSearch’ (264). Explicitly state that ‘Search-Intensive’ is the union of these two subsets to avoid ambiguity about whether it is a distinct category or a superset.
- **[writing]** The text in Section 4.1 claims a +7.0 gain for Phase 2 over the no-search DPO baseline. Verify that the values 56.9 (Phase 2) and 49.9 (baseline) in Table 3 are the correct figures to support this specific calculation, as the table layout separates these rows.

## Overreach

*Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_overreach.md

The paper makes several strong claims about the universality and recursive nature of its proposed “knowledge boundary” and co-training framework that exceed the empirical scope of the provided experiments.

First, the **Abstract** and **Conclusion** frame the work as establishing a foundation for “recursive self-improvement” and claim that the knowledge boundary is “discoverable” and “cannot disappear” regardless of scale. However, the experimental section (Section 5) and Table 1 present results from a **single iteration** of the co-training loop (one DPO pass followed by one RFT pass). The authors explicitly state in the **Appendix (Limitations)** that “multi-round co-training... could progressively sharpen this boundary,” admitting that the current results do not demonstrate recursion or convergence. The rhetoric of “recursive self-improvement” and the definitive claim that the boundary “cannot disappear” are not supported by the single-step evidence provided. To fix this, the authors should either narrow the claim to “a single-step co-training protocol” or provide empirical results from at least two or three iterative cycles to substantiate the “recursive” nature of the improvement.

Second, the **Conclusion** makes a universal claim about the limits of scaling: “No model, regardless of scale, can internalize knowledge... The knowledge boundary... cannot disappear.” This is a broad theoretical assertion about the fundamental limits of large language and vision models. The paper’s empirical evidence is limited to two open-weight generators (4B and 7B parameters) and a few commercial APIs. There is no data presented for larger models (e.g., 70B+ or frontier-scale models) where the proportion of “internalizable” knowledge might significantly increase, potentially challenging the claim that the boundary “cannot disappear.” While the intuition is plausible, the paper presents it as a demonstrated fact rather than a hypothesis. The authors should qualify this claim to reflect the tested scale range (e.g., “within the scales



we tested”) or cite external literature on scaling laws for knowledge retention to support the universal generalization.

Finally, **Finding 1** describes the performance collapse on search-intensive prompts as “universal” and the bottleneck as “structural” across all generators. While the data shows a consistent drop for the tested models, the term “universal” implies a property that holds for *any* possible model architecture or training regime. The study tests a specific subset of models. While the trend is strong, the absolute claim of universality is slightly overreaching given the limited sample of architectures. A more precise phrasing would be “observed across all tested generators” or “robust across diverse architectures,” reserving “universal” for a theoretical proof or a much broader empirical sweep.

These issues are primarily matters of **writing** and **framing**. The core contribution—a method to improve search-augmented generation via co-training—is well-supported by the single-step results. However, the paper’s rhetoric inflates the scope of these results to imply a solved, recursive, and universal phenomenon that the current data does not yet prove. Narrowing the claims to match the single-iteration, limited-scale evidence will align the paper’s confidence with its demonstrated scope.

#### Action items.

- **[writing]** Abstract claims ‘recursive self-improvement’ but experiments (Sec 5) show only one DPO/RFT pass. Appendix admits multi-round is future work. Change ‘recursive’ to ‘single-step’ or add multi-iteration results.
- **[writing]** Conclusion claims ‘No model, regardless of scale, can internalize knowledge’ universally. Experiments test only 4B/7B models. Narrow to ‘within tested scales’ or cite scaling law literature to support the universal claim.
- **[writing]** Finding 1 calls the bottleneck ‘universal’ across all generators. Data covers only a subset of models. Rephrase as ‘observed across all tested generators’ to avoid overgeneralizing beyond the empirical scope.

## Safety Ethics

*Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_safety\_ethics.md

This work presents a benchmark and co-training framework for agentic visual generation, focusing on the “world-knowledge bottleneck.” From a safety and ethics perspective, the paper is well-constructed and appropriately addresses the primary risks associated with its contributions.

The authors explicitly acknowledge the dual-use potential of improving visual generation for real-world entities (e.g., public figures, cultural symbols) in Appendix \ref{app:broader}, noting the risk of “misinformation” and “privacy concerns.” They propose concrete mitigations, including the exclusion of raw scraped images from the public release, the use of derived

attributes/metadata instead, and the inclusion of usage guidelines prohibiting malicious identity synthesis. This aligns with responsible release practices for generative AI research.

Regarding data provenance, the paper details that the dataset (\dataset{ }) is constructed from production user prompts and synthesized via template instantiation and LLM rewriting (Appendix \ref{app:benchmark}). The authors state that the seed entities are drawn from publicly available information and that the benchmark is designed for research evaluation. While the paper does not provide an IRB statement, this is appropriate as the work does not involve direct human subjects, sensitive personal data collection, or private user interactions; it analyzes aggregated, anonymized production logs and synthetic prompts. The “search corpus” consists of pre-executed, archived search sessions, and the authors commit to filtering out unsafe content (explicit sexual content, graphic violence, hateful material) before release (Appendix \ref{app:responsible\_release}).

The paper does not disclose operational details for cyber-offense or biohazard methods, nor does it present a system designed for covert surveillance or deception. The “naive search” failures discussed (concept corruption, copy effects) are technical limitations of the current models, not vulnerabilities to be exploited. The evaluation protocol uses an automated VLM judge, and the authors note the correlation with human ratings, avoiding the release of potentially biased or harmful human-annotated data.

Overall, the paper demonstrates a clear understanding of the ethical implications of its work, implements reasonable safeguards for data release, and avoids the specific high-risk patterns (unconsented human data, actionable exploit details, PII exposure) that would necessitate a revision or rejection. The risk profile is low and well-managed.

## Scientific Evidence

*Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_scientific\_evidence.md

The paper presents a compelling hypothesis regarding the “knowledge boundary” in agentic visual generation and proposes a co-training framework to discover it. However, the evidentiary support for the central claims relies on experimental designs that do not fully rule out alternative explanations such as random variance, baseline asymmetry, or confounding model scale.

First, the primary results in Table 1 (Section 4.2) and the progression in Figure 3 are reported as single-point estimates (e.g., “31.8” for Klein-4B Phase 2) with no indication of variance. In generative modeling, performance can fluctuate significantly across different random seeds due to the stochastic nature of diffusion sampling and DPO preference pair construction. A gain of 2.6 points (Phase 1 to Phase 2) is reported as a definitive improvement, but without standard deviations or confidence intervals from multiple seeds (e.g., 3-5 runs), it is impossible to determine if this effect is robust or merely a lucky initialization. The claim that the method produces “monotonic improvement” is currently unsupported by statistical evidence of stability.

Second, the comparison of the proposed method against baselines suffers from asymmetry in model capability and tuning. The “ReasonedSearch” baseline (Section 3.3) utilizes a frontier VLM (Gemini-3-Flash) for the gating decision, whereas the proposed method uses a fine-tuned 8B model. The “Oracle” baseline in Table 1 also uses this frontier model. The paper claims the 8B co-trained model “matches or exceeds” the Oracle, but this comparison conflates the benefit of the co-training protocol with the inherent superiority of the frontier model’s reasoning capabilities. A fair test of the *protocol* requires comparing the co-trained 8B model against a non-co-trained 8B model (or a similarly scaled baseline) to isolate the effect of the RFT calibration. Currently, the design cannot distinguish whether the gains come from the “teach-then-search” mechanism or simply from using a stronger reasoner than the naive baselines.

Finally, the “BlindSearch” baseline is defined as searching for every gap, while the proposed method learns to abstain. However, the “ReasonedSearch” baseline (which also abstains) is implemented with a much stronger model than the proposed method’s reasoner. The paper does not provide a control where the *same* 8B reasoner is used for both the baseline (SFT only) and the proposed method (SFT + RFT) to isolate the specific contribution of the rejection-sampling finetuning. Without this ablation, the claim that the *calibration* is the key driver of the “selectivity” (recovering NoSearch performance) remains plausible but not rigorously established.

To strengthen the evidence, the authors should report results with seed variance, ensure baselines use comparable model scales and tuning efforts, and include an ablation isolating the RFT phase from the SFT warm-up.

#### Action items.

- **[writing]** The paper presents a compelling hypothesis regarding the “knowledge boundary” in agentic visual generation and proposes a co-training framework to discover it. However, the evidentiary support for the central claims relies on experimental designs that do not fully rule out alternative explanations such as random variance, baseline asymmetry, or confounding model scale. First, the primary results in Table 1 (Section 4.2) and the progression in Figure 3 are reported as single-point estimates (e.g.

## Statistical Analysis

*Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_statistical\_analysis.md

The statistical treatment of the quantitative results in this paper is insufficient to support the strong inferential claims made regarding the efficacy of the co-training framework.

**1. Missing Uncertainty Reporting (Table 1 & Table 2)** The primary results in Table 1 (Main Results) and Table 2 (Naive Search) are presented as single point estimates (e.g., “31.8”, “76.5”, “56.9”) with no accompanying measure of variance (standard deviation, standard error, or confidence intervals). In the context of generative models, where results can vary significantly across different random seeds, training initializations, or even minor hyperparameter shifts,

reporting a single number creates an illusion of precision that is not supported by the data. The paper mentions using “multiple seeds” for some baselines in passing, but the main comparison tables do not reflect this. Without error bars or variance metrics, a reader cannot judge whether the reported “monotonic improvement” (e.g., from 29.2 to 31.8) is a robust effect or a result of random fluctuation. This is a reporting failure that obscures the reliability of the findings.

**2. Unsubstantiated Claims of Significance** The text frequently uses language implying statistical significance (e.g., “monotonic improvement,” “exceeding frontier generators,” “slightly exceeding the frontier oracle”) without providing the statistical machinery to back it up. For instance, the claim that the proposed method “slightly exceeds” the oracle (31.8 vs 31.2) in Table 1 is presented as a definitive finding. However, without a hypothesis test (such as a paired t-test or a bootstrap test given the likely non-normal distribution of scores) and a reported p-value, this difference is indistinguishable from noise. The paper fails to address the multiple comparisons problem inherent in testing across three phases, two architectures, and multiple difficulty sets (Set I, II, III, NoSearch). If 24 comparisons are made at  $\alpha = 0.05$ , one expects 1.2 false positives by chance alone. The absence of any correction (Bonferroni, Holm, or FDR) or explicit acknowledgment of this multiplicity undermines the validity of the “significant” gains claimed.

**3. Recommendation** To rectify these issues, the authors must:

1. Re-run the experiments (or aggregate existing runs if available) to report **mean  $\pm$  standard deviation** over at least 3 independent seeds for every number in Tables 1 and 2.
2. Perform appropriate **statistical hypothesis tests** (e.g., paired t-tests or Wilcoxon signed-rank tests) comparing the proposed method against the baselines for each stratum and architecture.
3. Apply a **multiple-comparison correction** (e.g., Holm-Bonferroni) across the set of all reported comparisons and report the adjusted p-values.
4. If re-running is not feasible, the text must be softened to describe “observed trends” rather than “statistically significant improvements,” and the tables must explicitly state that results are from a single run.

Currently, the numbers in the paper do not mean what the text claims they mean: they are presented as definitive, significant improvements when they are merely point estimates with unknown variance.

#### Action items.

- **[writing]** Tables 1 and 2 report single-point scores (e.g., 31.8) without uncertainty metrics (SD/SE/CI). Given the stochastic nature of generation, this implies false precision. Report mean  $\pm$  SD over 3 seeds for all results, or explicitly state results are from a single run and soften claims of ‘monotonic improvement’.
- **[science]** Section 5.1 claims ‘monotonic improvement’ and implies statistical significance without reporting p-values, confidence intervals, or hypothesis tests. With no variance reported, differences like 31.8 vs 31.2 are indistinguishable from noise. Perform paired tests

with multiple-comparison correction or rephrase claims as observed trends.

## Writing Quality

*Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_writing\_quality.md

The paper presents a compelling argument for agentic visual generation, but the prose occasionally falters in clarity and flow, requiring the reader to re-parse sentences or infer missing grammatical elements.

The most significant issue is the presence of sentence fragments and run-on sentences that disrupt the narrative momentum. In Section 1, the sentence “This renders an agentic visual generation problem: a generator is equipped with an agentic reasoner...” lacks a main verb for the subject “This,” forcing the reader to mentally insert a verb like “creates” or “defines.” Similarly, the transition “But we found that naive search fails” in the same section creates a comma splice when joined to the previous clause. These are not just stylistic choices; they are grammatical errors that break the reader’s immersion.

Paragraph structure also suffers from weak topic sentences. In Section 3.1, the description of the “Filter” stage begins with a passive construction (“After multimodal search is executed...”) rather than stating the stage’s purpose directly. A stronger opening like “The Filter stage selects references...” would immediately orient the reader. Additionally, the transition into the formal definition of the “Knowledge Boundary” in Section 3.2 is abrupt. The text jumps from a qualitative observation about shifting boundaries to a dense mathematical definition without a bridging sentence to explain *why* the formalization is necessary at that specific moment.

Finally, the abstract’s final sentence suffers from inconsistent parallelism in its list of released assets (“code, model and the full dataset, co-training corpus, and search corpus”), which makes the list difficult to parse quickly. While the scientific content is strong, these structural and grammatical friction points prevent the paper from reading as smoothly as it should. Addressing these specific sentence-level issues will significantly improve the reader’s ability to follow the argument without interruption.

### Action items.

- **[writing]** The paper presents a compelling argument for agentic visual generation, but the prose occasionally falters in clarity and flow, requiring the reader to re-parse sentences or infer missing grammatical elements. The most significant issue is the presence of sentence fragments and run-on sentences that disrupt the narrative momentum. In Section 1, the sentence “This renders an agentic visual generation problem: a generator is equipped with an agentic reasoner...” lacks a main verb for the subject “